



RS-485 Communications
Interface Control Definition
For Ez-Link HMI

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Revision Status

Rev	Date	Author	Company	Description
A	05/29/24	M. Gonzalez	Nidec Drive Systems	Initial version
B	07/08/24	T. Blazier	Nidec Drive Systems	Revised using KC packet within Pentair packet design
C	07/11/24	T. Blazier	Nidec Drive Systems	• Removed "MC Cal Version" from Table 32 • Adjusted range of Run command modes in Table 8 • Removed obsolete Run command modes in Table 9 • Removed unused "Command Mode" in Table 28 • Removed obsolete "Operating Mode" in Table 28 • Removed obsolete "Motor Run mode" in Table 28 • Removed obsolete "Motor Run command" in Table 28 • Updated Message Length in Table 28 • Removed obsolete Table 31 • Section 3.5.5 changed all "DI" references to "HMI" • Fixed mislabeled byte indexes in some tables
D	07/23/24	T. Blazier	Nidec Drive Systems	• Updated Table 23 with three additional HMI modes • Added Schema row to Table 28 and adjust data length • Removed EE CRC from Table 31

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1.0 INTRODUCTION

1.1 Purpose

The purpose of this document is to define the RS-485 communications port interface for the EZ-Link communication protocol HMI boards. The RS-485 communications port is used for production testing and diagnostics. During normal operation the RS-485 interface for testing and diagnostic purposes is inactive. This document details the communication interface for production testing only.

The intended audience for this document is any software developer(s) working on the project. This document may also be reviewed by external regulatory entities to ensure standards conformance. The document is labeled confidential and is considered company proprietary material.

1.2 Background

The control module consists of two printed circuit boards (PCBs); a motor control board (often abbreviated as MC) and a Human Machine Interface (HMI) board. Each board contains a microprocessor which executes software to implement the required functionality. The motor control board is connected to the AC line inputs and the motor. It is responsible for generating onboard voltages from the AC line input and controlling the 3-phase magnet motor. The HMI control board is integrated into the top cover keypad assembly and is responsible for interfacing with a user and issuing commands to the motor control board. The two control boards communicate with each other through a dedicated RS-232 serial interface and communicate via the Ez-Link protocol. The RS-485 port is a separate communications interface and is connected only to the HMI control board. The RS-485 port is the focus of this document, Ez-Link and the RS-232 interface between control board are covered in a separate specification.

1.3 Document Overview

This document contains the interface control definition for the Variable Speed Motor RS-485 communications port including the hardware signaling and communications protocol. This Interface Control Definition (ICD) is organized as follows:

Section	Purpose
1	Document purpose, background, and document overview
2	RS-485 communications port hardware interface
3	RS-485 communications port serial interface

Table 1 : Document Overview

This ICD is an evolutionary document and may be modified during development to reflect changes in the plan. Any modifications to this document will be noted in the Revision Status table. This is a controlled document and changes will follow the process outlined in the Software Configuration Management (SCM) Plan.

1.4 Definitions, Acronyms, and Abbreviations

The following terms, acronyms, and abbreviations used in this document are defined below so that this document can be properly interpreted.

Human Machine Interface (HMI): The system by which people interact with a machine. The human machine interface includes hardware (physical) and software (logical) components.

Software Configuration Management (SCM): The task of tracking and controlling changes in the software and related documents including revision control and the establishment of baselines.

Interface Control Definition (ICD): Document containing a detailed definition of the data format, protocol, and timing requirements for an interface to an external entity.

1.5 Reference Documents

This document shall be used in conjunction with the following publications:

- KCSC0231 schematic
- Pentair RS485 VS Drive Communications Specification Version 0.04

2.0 HARDWARE INTERFACE

The HMI RS-485 communications port is located on the external connector attached to the side of the control module's top cover. This port is connected directly to the HMI control board. The pin definitions and assignments for the external connector are as follows:

Pin	Definition	Signal Range
1	+5V Supply *	0 – 20ma
2	RS-485 A (Data -)	0 – 5V
3	RS-485 B (Data +)	0 – 5V
4	Schedule 1 Digital Control	0, 5 - 30V AC/DC
5	Schedule 2 Digital Control	0, 5 - 30V AC/DC
6	Schedule 3 Digital Control	0, 5 - 30V AC/DC
7	Quick Clean Digital Control	0, 5 - 30V AC/DC
8	5V *, RS-485 Common Ground	0V

* – For control systems with flow control, +5V is replaced with +24V

Table 2 : External Connector Pin Definition

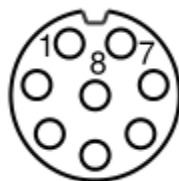


Figure 1 : External Connector Pin Assignments

All HMI devices contain a reference 5V supply that is used to power the RS-485 interface on the PCB. HMIs that are not configured to connect to an external flow control device are also able to use this same 5V supply to control the digital inputs. For these HMIs, +5V is present on pin 1 of the connector. HMIs that are configured to connect to an external flow control device have a separate 24V supply on the PCB which can be used to power an external flow sensor and/or control the digital inputs. For these HMIs, +24V is present on pin 1 of the connector.

Note that vendors of RS485 devices do not always follow the standard when assigning labels to the differential data pair A/B (-/+). In this case the data pair connections will need to be reversed to communicate with the HMI control board.

The RS-485 port operates using half duplex signaling. Collision detection can be implemented at the data link layer and/or handled by the packet protocol. The port settings are fixed and specified in Table 3 – there is no automatic baud rate detection.

Port Setting	Value
Baud Rate	9600 bps
Data Bits	8 (lsb first)
Parity	None
Start Bits	1
Stop Bits	1
Flow Control	None

Table 3 : Port Configuration

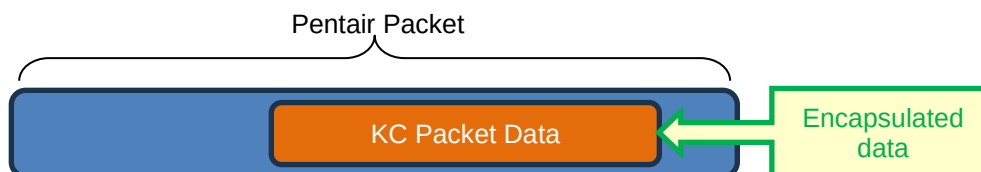
3.0 SERIAL INTERFACE

For purposes of factory testing, an external testing device such as a PC is connected to the HMI. This external device issues requests to the HMI to obtain data or to elicit a response from the control. The control then issues a reply which may contain data or simply an acknowledgment that the request was received.

In earlier HMI controls, a separate protocol was used for factory testing. This protocol (the Kinetek KC protocol) ran concurrently with the Pentair protocol on the same RS-485 line and the coexistence of the two occasionally created conflicts. Within the **Pentair RS485 VS Drive Communications Specification** there is an application note copied here:

The Drive may not actively support any 'unprotected' protocols while supporting the VS Drive protocol(s). Since the drive will see all of the Automation link traffic, virtually any combination of data bytes are possible. This requires that any active protocol uses sync bytes, unit addresses, checksums and the like to ensure no unexpected drive actions occur from unrelated data on the link

Maintaining two protocols is in violation of this directive, therefore a hybrid solution is implemented to encapsulate the factory data (previously used in the KC packets) within a Pentair packet:



3.1 Pentair Protocol Overview

Because all KC packets will be "inside" a Pentair packet, the overall structure of the Pentair packet format needs to be understood first. The basic structure of a Pentair packet is shown in Table 4 – this is just an overview of the bytes and their purpose. For complete details of the packet bytes and the protocol in general, review the **Pentair RS485 VS Drive Communications Specification**.

Packet Byte	Value	Description
1	0xFF	Preamble Byte 1
2	0x00	Preamble Byte 2
3	0xFF	Preamble Byte 3
4	0xA5	Sync Byte
5	0x00 – 0xFF	House address
6	0x00 – 0xFF	Address of receiver
7	0x00 – 0xFF	Address of sender
8	0x0D	Operation code (Nidec reserved op code)
9	0x00 – 0x30	Information field length (N, maximum length = 40)
10 to (N+9)	0x00 – 0xFF	Information bytes (if N != 0)
N + 10	0x00 – 0xFF	Additive checksum of bytes 4 through N + 9 (bits 8 – 15)
N + 11	0x00 – 0xFF	Additive checksum of bytes 4 through N + 9 (bits 0 – 7)

Table 4 : Pentair Packet Protocol

The encapsulated, legacy KC data will be transmitted in the "Information bytes" section of the Pentair packet. The byte ordering of the KC data is shown in Table 5:

KC Packet Byte	Pentair Packet Byte	Value or Range	Description
1	10	0xD0	Packet Type (0xD0 = Factory Test Command)
2	11	0x51 – 0x5E	Command Message Number
		0x71 – 0x77	Response Message Number
3	12	0x00 – 0x40	Size of Message Data Bytes
4 to N *	13 to (N+9) *	0x00 – 0xFF	Message Data Bytes

* – 'N' is value from Table 4

Table 5 : KC Packet Data

Example

Here is what a complete Pentair packet with encapsulated KC factory data looks like:

Preamble & Sync				House	Dest	Source	Op Code	Length	Type	Command	Size	Data	Checksum	
FF	00	FF	A5	00	60	10	0D	04	D0	51	01	01	02	49

The **Preamble & Sync** bytes are fixed. For variable speed pumps, the **House** address is always 0x00. Per the **Pentair RS485 VS Drive Communications Specification**, the **destination** address can be 0x10 for the pool control system, 0x60 – 0x6F for the VS drive or 0xF for a broadcast value. Similarly, the **source** address can be 0x10 for the pool control system or 0x60 – 0x6F for the VS drive. The **Op Code** is always 0xD for Nidec specific commands and the **Type** is always 0xD0 for factory test commands. The **length** field ('N') will vary depending on the encapsulated KC command data as will the number of items in the "Information Bytes" field. In this example, the KC **command** is 0x51 (Command Mode) and the **size** (or count) of parameter bytes is 1 and the single parameter byte (**Data**) is 1. Finally, $A5 + 00 + 60 + 10 + 0D + 04 + D0 + 51 + 01 + 01 = 0x249$.

3.2 Message Types

There are two types of messages defined for this interface: commands and responses. Command messages are sent to the HMI and response messages are sent from the HMI. When the HMI receives a valid command message it transmits a response message back to the sender. The response message is an acknowledgement that the command packet was received correctly and contains any requested information. If no response to the command message is defined or there was an issue with the received message, the HMI will respond with an Acknowledge response message.

The Acknowledge response message contains a status byte which indicates if the received message data is valid, invalid or the requested data is unavailable. The Acknowledge response message is only sent for correctly received command packets. The HMI will not respond to incorrectly formatted or received packets. The sender usually retransmits a command message when it is not acknowledged after some period of time.

3.3 Communication Modes

There are two modes of RS-485 communication with the HMI: production test and firmware download. This document only details the production test mode. This mode is used during manufacturing to validate new HMI controls, to validate a complete motor and control assembly and to analyze product returns for issues.

3.4 Command Messages

Command messages are sent to the HMI to command an action or request data. Each command message is encapsulated within an outer Pentair packet. The complete command message set is shown in Table 6 followed by a detailed description of each command.

Command Message	ID	Description
Command Mode	0x51	Operate in production test or monitor mode
Run Command	0x52	Operate in a specified Run mode
Read Motor Status	0x53	Request Motor Control status and operating data
Firmware Version Query	0x56	Request version of motor control and HMI Firmware
Reset Microprocessor	0x5A	Perform core reset of HMI and/or motor control microprocessors
Control LEDs	0x5B	Activate/Deactivate each individual keypad LED
Control LCD Backlight	0x5C	Control brightness of LCD backlight
Write LCD Text	0x5D	Write alphanumeric text to LCD
Control LCD Icons	0x5E	Activate/Deactivate individual icons on LCD

Table 6 : Serial Command Messages

3.4.1 Command Mode

The Command Mode message instructs the control system to operate in either monitor or production test mode. Monitor mode is the normal default operating mode where the motor operates based on settings programmed into the HMI, external digital inputs or manually from the keypad. In monitor mode an external entity can request status information from the HMI through the RS-485 interface but not affect operation.

The purpose of production test mode is to enable all aspects of the motor to be tested and verified during the manufacturing process. In production test mode the motor operates according to commands received from the RS-485 interface. The motor ignores all operational commands from programmed settings, external digital inputs, and keypad inputs. The HMI will enter production test mode if a Command Mode message with command mode set to Production Test is received through the RS-485 interface.

To maintain the HMI in production test mode the Command Mode message must be periodically sent at intervals of less than 30 seconds. In production mode the Command message serves as a safety interlock to prevent the motor from operating without regular direct intervention. The HMI will return to monitor mode if a Command mode message is not received within 30 seconds. When entering or exiting production test mode the HMI will terminate any current operation and place the motor in the OFF state.

An HMI Status message is sent in response to the Command Mode message. The format of the Command Mode message follows:

Message Byte	Value or Range	Description
1	0x51	Message ID
2	1	Message data length
3	0x00 – 0x01	Command mode (0 = Monitor, 1 = Production Test)

Table 7 : Command Mode Message Format

3.4.2 Run Command

The Run Command message instructs the HMI to execute one of the run commands listed in Table 9. For the Run Command message to be processed the control must be in production test mode. There is no response to this command message. The run command status can be verified by sending a Command Mode command message and reading the HMI Status response message. The format of the Run Command message follows:

Message Byte	Value or Range	Description
1	0x52	Message ID
2	4	Message data length
3	0x00 – 0x01	Run command (see Table 9)
4	0x00 – 0x01	Direction (0 = CW, 1 = CCW; facing shaft end of motor)
5	0x00 – 0xFF	Target speed (RPM) (bits 8-15) (450 ≤ RPM ≤ 3450)
6	0x00 – 0xFF	Target speed (RPM) (bits 0-7) (450 ≤ RPM ≤ 3450)

Table 8 : Run Command Message Format

ID	Command/Mode	Description
0	Off	Decelerate motor to stop
1	Run	Run motor to speed

Table 9 : Run Modes/Commands

3.4.3 Read Motor Control Status

The Read Motor Control Status command instructs the HMI to read and return the status of the motor control board. The HMI reads the motor control status and replies with a Motor Control Status message when this command message is received. The format of the Read Motor Control Status command message follows:

Message Byte	Value or Range	Description
1	0x53	Message ID
2	0	Message data length

Table 10 : Read Motor Control Status Message Format

3.4.4 Firmware Version Query

The Version Query command instructs the HMI to return version information for the HMI and Motor Control firmware. The format of the Firmware Version Query command message is shown in Table 11. A Firmware Version Response message is sent as a reply to this command message.

Message Byte	Value or Range	Description
1	0x56	Message ID
2	0	Message data length

Table 11 : Firmware Version Query Message Format

3.4.5 Reset Microprocessor

The Reset Microprocessor command instructs the HMI to generate a core reset for the HMI or Motor Control microprocessor. This command is executed regardless of the motor operating state. **To be safe the motor should be stopped before this command is issued.** There is no response to this command message. This command is only available in production test mode. The format of Reset Microprocessor command message follows:

Message Byte	Value or Range	Description
1	0x5A	Message ID
2	1	Message data length
3	0x01 – 0x02	Microprocessor Selection (1 = HMI, 2 = Motor Control)

Table 12 : Reset Microprocessor Message Format

3.4.6 Control LEDs

The Control LEDs command instructs the HMI to separately activate or deactivate each individual LED on the user keypad. There is no response to this command message. This command is only available in production test mode. The format of Control LEDs command message follows:

Message Byte	Value or Range	Description
1	0x5B	Message ID
2	2	Message data length
3	0x00 – 0x03	LED control word (0 = Off, 1 = On) (bits 8-15) (see Table 14)
4	0x00 – 0xFF	LED control word (0 = Off, 1 = On) (bits 0-7)

Table 13 : Control LEDs Message Format

LED Function	Bit Mask
Speed	0x001
Time	0x002
Duration	0x004
Watts	0x008
Schedule 1	0x010
Schedule 2	0x020
Schedule 3	0x040
Quick Clean	0x080
Start/Stop	0x100
Digital Control	0x200

Table 14 : LED Control Word Definition

3.4.7 Control LED Backlight

The Control LED Backlight command instructs the HMI to adjust the brightness of the LCD backlight on the user keypad. There is no response to this command message, but the LCD brightness will take effect immediately. This command is only available in production test mode. The format of Control LED Backlight command message follows:

Message Byte	Value or Range	Description
1	0x5C	Message ID
2	1	Message data length
3	0x00 – 0x64	LCD brightness level (0 = Off, 100 = Maximum)

Table 15 : Control LED Backlight Message Format

3.4.8 Write LCD Text

The Write LCD Text command instructs the HMI to display a text string on the LCD panel. Only characters defined in a standard 7 segment LCD font map can be displayed. Unprintable characters will be displayed as blank characters. There is no response to this command message. This command is only available in production test mode. The format of Write LCD Text command message follows:

Message Byte	Value or Range	Description
1	0x5D	Message ID
2	4	Message data length
3	'0'-'9','A'-'Y','b'-'u'	LCD character 1 (leftmost)
4	'0'-'9','A'-'Y','b'-'u'	LCD character 2
5	'0'-'9','A'-'Y','b'-'u'	LCD character 3
6	'0'-'9','A'-'Y','b'-'u'	LCD character 4 (rightmost)

Table 16 : Write LCD Text Message Format

3.4.9 Control LCD Icons

The Control LCD Icons command instructs the HMI to individually display or hide icons on the LCD panel. There are 6 icons on the LCD panel: 3 periods, a colon, an |, and a PM. The | icon is positioned below the outer arc of the P icon so that the concurrent display of | and PM forms an AM icon. This command does not affect the 7 segment characters. There is no response to this command message. This command is only available in production test mode. The format of Control LCD Icons command message follows:

Message Byte	Value or Range	Description
1	0x5E	Message ID
2	1	Message data length
3	0x00 – 0x1F	LCD icon control word (0 = Off, 1 = On)(see Table 18)

Table 17 : Control LCD Icons Message Format

LCD Icon	Bit Mask
Period 1 (between character 1 and 2)	0x001
Period 2 (between character 2 and 3)	0x002
Period 3 (between character 3 and 4)	0x004
Colon	0x008
PM	0x010
(requires PM icon to create AM icon)	0x020

Table 18 : LCD Icon Control Word Definition

3.5 Response Messages

Response messages are messages sent from the HMI after a command message is received, validated, and processed. In general, the HMI only sends messages in response to a command message. Not all command messages generate responses. When a command message does not generate a specific response message an Acknowledge message is returned to acknowledge message reception. The HMI response messages are shown in Table 19 followed by a detailed description of each message.

Response Message	ID	Description
Acknowledge	0x71	Used to indicate active link and returns message status info
HMI Status	0x72	Status and sampled values for the HMI Control
Motor Control Status	0x73	Status for the motor control drive, control, and fault monitor
Firmware Version Response	0x76	Version and checksum info for HMI and Motor Control firmware

Table 19 : HMI Response Messages

3.5.1 Acknowledge

The Acknowledge message is sent to acknowledge reception of the last received command message. Normally reception of a command message is acknowledged in a command's response message. The Acknowledge message is sent if the command message has no defined response message, the response data is unavailable, or the message contents are invalid. The purpose of the Acknowledge message is stored in the status flag (see Table 21). The format of the Acknowledge message follows:

Message Byte	Value or Range	Description
1	0x71	Message ID
2	1	Message data length
3	0 – 3	Message Status (see Table 21)

Table 20 : Acknowledge Message Format

Message Status Code	Description
0	Message is valid, no defined response
1	Message ID or length invalid
2	—
3	Response message data is unavailable

Table 21 : Message Status Definition

3.5.2 HMI Status

The HMI Status message is sent as a response to the Command Mode command message. This message contains the complete status from the HMI control. The format of the HMI Status message follows:

Message Byte	Value or Range	Description
1	0x72	Message ID
2	8	Message data length
3	0x00 – 0xFF	Keypad Status (see Table 24)
4	0x00 – 0x0F	Digital Input Status (see Table 26)
5	0x00 – 0x08	HMI Mode (see Table 23)
6	0x00 – 0x03	HMI Sub-mode (see Table 25)
7	0x00 – 0x05	Total minutes since midnight (bits 8-15)
8	0x00 – 0xFF	Total minutes since midnight (bits 0-7)
9	0x00 – 0x3B	Partial minutes (seconds)
10	0x00 – 0xFF	HMI Fault Status (see Table 27)

Table 22 : HMI Status Message Format

HMI Mode	ID
Stop	0
Clock Setup	1
Schedule Setup	2
Start	3
Flow Setup	4
Override	5
Quick Clean Override	6
Digital Control	7
Automation Control	8
Service	9
Production Test	10
HMI Setup	11

Table 23 : HMI Mode Definition

Keypad Key	Bit Mask
Schedule 1	0x01
Schedule 2	0x02
Schedule 3	0x04
Quick Clean	0x08
Start/Stop	0x10
Up Arrow	0x20
Down Arrow	0x40
Display	0x80

Table 24 : Keypad Status Definition

HMI Sub-mode	ID
Normal	0
Prime	1
Fault	2
Lock Status Display	3

Table 25 : HMI Sub-mode Definition

Digital Input	Bit Mask
Schedule 1	0x01
Schedule 2	0x02
Schedule 3	0x04
Quick Clean	0x08

Table 26 : Digital Input Status Definition

Fault Status	Code
No Fault	0x00
Com link failure (HMI side)	0x21

Table 27 : Fault Status Definition

3.5.4 Motor Control Status

The Motor Control Status message is sent as a response to the Read Motor Control Status command message. This message contains the combined status for the drive, control, temperature and fault monitors. The format of the Motor Control Status message is shown in Table 28.

Message Byte	Value or Range	Description
1	0x73	Message ID
2	34	Message data length
3	0x00 – 0xFF	Report schema
4	0x00 – 0xFF	Fault status (bits 24-31) (see bit assignments in Table 29)
5	0x00 – 0xFF	Fault status (bits 16-23) (see bit assignments in Table 29)
6	0x00 – 0xFF	Fault status (bits 8-15) (see bit assignments in Table 29)
7	0x00 – 0xFF	Fault status (bits 0-7) (see bit assignments in Table 29)
8	0x00 – 0x04	MCU state (see Table 30)
9	0x00 – 0x01	Stopped state (0 = Motor is not stopped, 1 = Motor is stopped)
10	0x00 – 0xFF	Module temperature (°C x 10) (bits 8-15)
11	0x00 – 0xFF	Module temperature (°C x 10) (bits 0-7)
12	0x00 – 0xFF	PFC temperature (°C x 10) (bits 8-15)
13	0x00 – 0xFF	PFC temperature (°C x 10) (bits 0-7)
14	0x00 – 0xFF	Bridge temperature (°C x 10) (bits 8-15)
15	0x00 – 0xFF	Bridge temperature (°C x 10) (bits 0-7)
16	0x00 – 0xFF	AC line voltage (RMS volts) (bits 8-15)
17	0x00 – 0xFF	AC line voltage (RMS volts) (bits 0-7)
18	0x00 – 0x01	AC line voltage state (0 = 240 VAC, 1 = 120 VAC)
19	0x00 – 0xFF	Bus voltage (DC volts) (bits 8-15)
20	0x00 – 0xFF	Bus voltage (DC volts) (bits 0-7)
21	0x00 – 0xFF	Control's estimated input power (Watts) (bits 8-15)
22	0x00 – 0xFF	Control's estimated input power (Watts) (bits 0-7)
23	0x00 – 0xFF	Commanded speed (bits 8-15)
24	0x00 – 0xFF	Commanded speed (bits 0-7)
25	0x00 – 0xFF	Target shaft speed (bits 8-15)
26	0x00 – 0xFF	Target shaft speed (bits 0-7)
27	0x00 – 0xFF	Motor phase current (Amps x 1000) (bits 8-15)
28	0x00 – 0xFF	Motor phase current (Amps x 1000) (bits 0-7)
29	0x00 – 0xFF	Motor line-to-line input voltage (RMS) (bits 8-15)
30	0x00 – 0xFF	Motor line-to-line input voltage (RMS) (bits 0-7)
31	0x00 – 0xFF	Motor input power (Watts) (bits 8-15)
32	0x00 – 0xFF	Motor input power (Watts) (bits 0-7)
33	0x00 – 0xFF	Shaft torque (N•m x 100) (bits 8-15)
34	0x00 – 0xFF	Shaft torque (N•m x 100) (bits 0-7)
35	0x00 – 0xFF	Horsepower (HP x 100) (bits 8-15)
36	0x00 – 0xFF	Horsepower (HP x 100) (bits 0-7)

Table 28 : Memory Read Response Message Format

Bit	Description
0	Power up reset
1	Watchdog reset
2	Motor stalled
3	Hardware over current
4	AC line under voltage
5	Comm timeout
6	Under voltage
7	Over voltage
8	Bridge hot
9	Inverter hot
10	Software over current
11	Open phase
12	Open and shorts test failed
13	PFC hot (or cold)
14	EEPROM CRC failed
15	Reserved
16	Inverter temperature exceeded limit backdown
17	PFC temperature exceeded limit backdown
18	Bridge temperature exceeded limit backdown
19	Torque load exceeded limit backdown
20	D vector current exceeded limit backdown
21	Motor current exceeded limit backdown
22-31	Unused event bits

Table 29 : Fault Report Bit Assignments

MCU State	ID
Reset state	0
Pre-charge delay state	1
Stop state	2
Fault state	3
Run state	4

Table 30 : MCU State Definition

3.5.5 Firmware Version Response

The Firmware Version Response message is sent as a response to a Firmware Version Query command message. This message contains version information for the HMI and Motor Control Firmware. The format of the FW Version Response message is found in Table 31. (**HMI**=Human Machine Interface, **MC**=Motor Control)

Message Byte	Value or Range	Description
1	0x76	Message ID
2	28	Message data length
3	0x00 – 0xFF	HMI Bootloader Version (bits 8-15)
4	0x00 – 0xFF	HMI Bootloader Version (bits 0-7)
5	0x00 – 0xFF	HMI Bootloader CRC (bits 8-15)
6	0x00 – 0xFF	HMI Bootloader CRC (bits 0-7)
7	0x00 – 0xFF	HMI Application Version (bits 8-15)
8	0x00 – 0xFF	HMI Application Version (bits 0-7)
9	0x00 – 0xFF	HMI Application CRC (bits 8-15)
10	0x00 – 0xFF	HMI Application CRC (bits 0-7)
11	0x00 – 0xFF	HMI EE Version (bits 8-15)
12	0x00 – 0xFF	HMI EE Version (bits 0-7)
13	0x00 – 0xFF	MC Drive Type (bits 8-15)
14	0x00 – 0xFF	MC Drive Type (bits 0-7)
15	0x00 – 0xFF	MC Motor Type (bits 8-15)
16	0x00 – 0xFF	MC Motor Type (bits 0-7)
17	0x00 – 0xFF	MC UL Version (bits 8-15)
18	0x00 – 0xFF	MC UL Version (bits 0-7)
19	0x00 – 0xFF	MC Bootloader Version (bits 8-15)
20	0x00 – 0xFF	MC Bootloader Version (bits 0-7)
21	0x00 – 0xFF	MC Bootloader CRC (bits 8-15)
22	0x00 – 0xFF	MC Bootloader CRC (bits 0-7)
23	0x00 – 0xFF	MC EE Version (bits 8-15)
24	0x00 – 0xFF	MC EE Version (bits 0-7)
25	0x00 – 0xFF	MC EE CRC (bits 8-15)
26	0x00 – 0xFF	MC EE CRC (bits 0-7)
27	0x00 – 0xFF	MC Application CRC (bits 8-15)
28	0x00 – 0xFF	MC Application CRC (bits 0-7)
29	0x00 – 0xFF	MC Application Version (bits 8-15)
30	0x00 – 0xFF	MC Application Version (bits 0-7)

Table 31 : Firmware Version Response Message Format